14 Cone **A** and cone **B** are mathematically similar.
The ratio of the volume of cone **A** to the volume of cone **B** is 27 : 8

The surface area of cone **A** is 297 cm^2

Show that the surface area of cone **B** is 132 cm^2

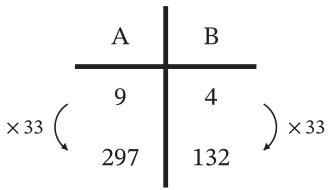
Volume ratio A : B = 27 : 8

Length ratio = 3 : 2 (cube roots)

1 mark
Cube-root the volume ratio

Area ratio = $3^2 : 2^2 = 9 : 4$

1 mark
Square the length ratio



Area of B = 297 ÷ 9 × 4 = 132 cm^2

1 mark
Required area shown from the correct ratio

(Total for Question 14 is 3 marks)